IS regular course 2025-2026

Chapter 1 – Introduction to Science

Distinguish variables

<u>Independent variable and dependent variables 自變數與應變數</u>

<u>Example1 : Does the Weight of McDonald's Burgers Decrease Over Time?</u>

- In this question, we are actually finding **relationship** 關係 between to **variables 變數:**

比影響	Affects 影響	被影響
Time	→	**Weight of McDonalds burgers**

In other words, the relationship between

independent variable 自變數 and **dependent variables 應變數**

比影響	Affects 影響	被影響
Independent variable ★ONLY ONE!★	→	**Dependent variable**

So, the **independent variable** in this experiment is [time / weight of Mcdonalds burgers]
the **dependent variable** in this experiment is [time / weight of Mcdonalds burgers]

IS regular course 2025-2026

1. Does Listening to Music While Studying Improve Focus?
學習時聽音樂是否能提高專注力？

Independent variable	Presence of music while studying / Level of focus
Dependent variable	Presence of music while studying / Level of focus

2. Does the Use of Social Media Affect Study Efficiency?
使用社交媒體是否會影響學習效率？

Independent variable	Time spent on social media / Study efficiency
Dependent variable	Time spent on social media / Study efficiency

3. Does the Amount of Screen Time Affect Eye Strain?
屏幕使用時間是否會影響眼睛疲勞？

Independent variable	Duration of screen time / Level of eye strain
Dependent variable	Duration of screen time / Level of eye strain

4. Does the Frequency of Hand Washing Reduce the Spread of Germs?
洗手頻率是否能減少細菌傳播？

Independent variable	
Dependent variable	

5. Does the Amount of Study Time Affect Test Scores?
學習時間是否會影響考試成績？

___________ variable	Study time
___________ variable	Test scores

6. What is the Relationship Between Study Environment Noise Level and Concentration?
學習環境的噪音水平與專注力之間的關係是什麼？

Independent variable	
Dependent variable	

Fair test 公平測試 and control variables 控制變數

- A fair test in science ensures that **only one variable is changed** at a time while all other conditions are kept constant

It's nice to have a cold drink in the summer, but once we take it out of the refrigerator, it starts to become warmer

- A student find that **the rate of cold drink become warmer** in different cup are different, and it may be due to the **material of the cup**

Independent variable	
Dependent variable	

- We can prepare the following setup, pour the cold drink to different cups, and measure the time takes for the drinks become room temperature

- **Think：** 1. if the size of the cups are different, is the experiment fair?
  2. If the initial temperature of drinks in metal cup, ceramics cup plastic cup are 0℃, 8℃, 30℃ respectively, is the experiment fair?

So, in a **fair test**,
only one variable is changed, it is Independent variable !!!
Dependent variable is the result
Any other variable should be the same, which is control variables !!!

Example2 : Does the material of cup affect heating of drinks?

Example 3: Does the Daily Number of Steps Increase with a Fitness Tracker?

Variables	
Independent variable 自變數 (only one)	**Presence of fitness tracker**
Dependent variable 應變數	**Daily number of steps**
Controlled variables 控制變數	

Which of the following are **appropriate** controlled variables of this investigation?

A. The age and fitness level of participants.
參與者的年齡和健康水平

B. The brand and model of the fitness tracker used.
使用的健身追蹤器的品牌和型號

C. The planet on which the investigation takes place.
實驗進行的星球

D. The color and pattern on shoes.
鞋子的顏色和圖案

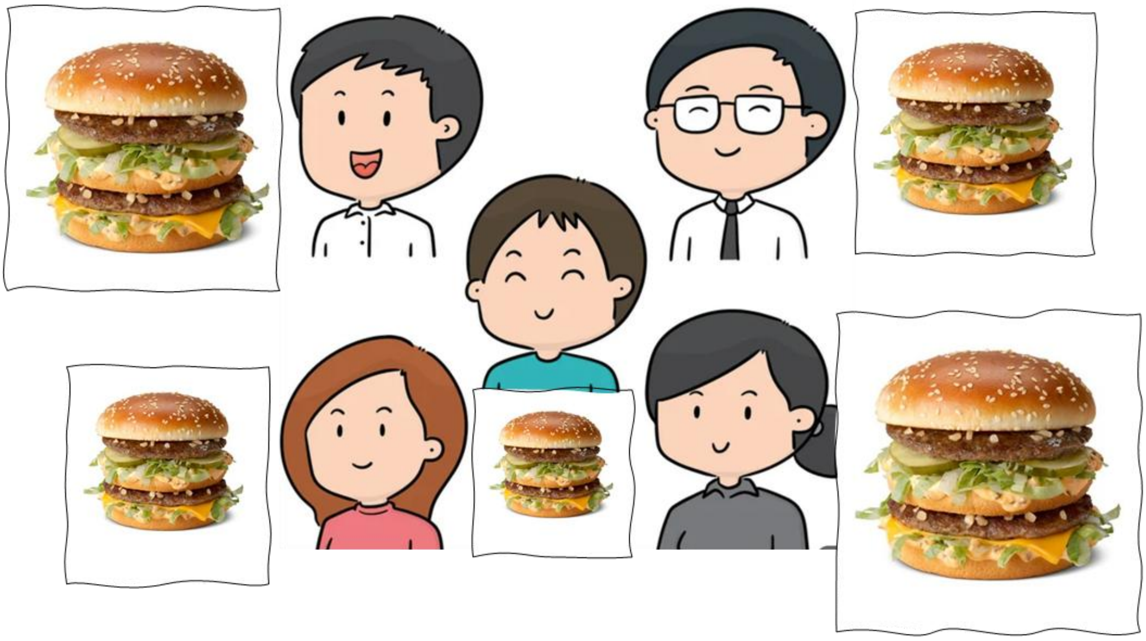
E. The time of day when steps are recorded.
記錄步數的時間

F. The location where participants walk (e.g., indoors or outdoors).
參與者行走的位置（例如室內或室外）

Scenario based question

Scenario 情境描述	Independent Variable 自變數	Dependent Variable 應變數	Controlled Variables 控制變數
1. A chef wants to know if the thickness of pizza dough affects how crispy the crust becomes after baking. 一位廚師想知道比薩麵團的厚度是否會影響烘烤後餅皮的酥脆程度。			
2. A student investigates whether the number of practice sessions per week affects the speed at which they learn a new song on the piano. 學生研究每週練習次數是否會影響學習新鋼琴曲的速度。			
3. A gardener wants to know if watering plants at different times of day affects the number of flowers produced. 園丁想知道在一天中不同時間澆水是否會影響開花數量。			
4. A class examines whether the size of paper airplanes affects how far they fly. 班級研究紙飛機大小是否會影響飛行距離。			
5. A group of friends test if the temperature of water used to brew tea affects the taste rating given by tasters. 一群朋友測試泡茶用水溫度是否會影響品嚐者的口味評分。			

Reliability 可靠性



If we only measure a single sample, there would be **experimental errors 實驗誤差** because there are **individual differences 個體差異.**

We can reduce these errors by

★Repeating 重複 measurement/experiment and **★taking average 平均**

300 g　　150g　　320g　　280g　　130g

Averaging

$$average\ weight = \frac{300 + 150 + 320 + 280 + 130}{5} = 236g$$

IS regular course 2025-2026

A. MCQs

1. Why is it important to take multiple measurements in an experiment?

A. To save time

B. To reduce experimental errors caused by individual differences

C. To increase the weight of the sample

D. To avoid performing calculations

2. What happens if we do not repeat measurements and take averages in an experiment?

A. The results will be more reliable.

B. Experimental errors caused by individual differences may affect the accuracy 準確性.

C. The total weight will increase.

D. The experiment will automatically become valid 有效的.

B. Short questions

1. A student measured the weights of three burgers as 300 g, 320 g, and 280 g. The student claims the average weight is 320 g.

   Question: Is the student correct? Show your calculations to verify the claim. (3 marks)

   ____

   ____

   ____

2. A group of students recorded the number of pages they read in a week: 45, 60, 55, 40, and 50 pages.
   What is the average number of pages read by the students?

   ____

   ____

   ____

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Answer Key

Fair test 公平測試 and control variables 控制變數

Scenario	Independent Variable	Dependent Variable	Controlled Variables
1	Thickness of pizza dough 比薩麵團的厚度	Crispiness of the crust after baking 烘烤後餅皮的酥脆程度	Baking time, oven temperature, type of flour, amount of toppings 烘烤時間、烤箱溫度、麵粉種類、配料用量
2	Number of practice sessions per week 每週練習次數	Speed of learning a new song 學習新曲目的速度	Song difficulty, length of each session, piano quality, prior experience 曲目難度、每次練習時長、鋼琴品質、先前經驗
3	Time of day for watering 澆水的時間	Number of flowers produced 開花數量	Plant species, amount of water, soil type, sunlight exposure 植物品種、澆水量、土壤種類、日照時間
4	Size of paper airplane 紙飛機的大小	Distance flown 飛行距離	Paper type, folding method, throwing force/angle, wind conditions 紙張種類、摺法、投擲力度/角度、風速風向
5	Temperature of water used to brew tea 泡茶用水的溫度	Taste rating given by tasters 品嚐者的口味評分	Type of tea, brewing time, amount of tea leaves, cup type 茶葉種類、沖泡時間、茶葉用量、杯子種類

Reliability 可靠性

A. MCQs

1. B 2. B

B. Short questions

1.

Total weight = $300 + 320 + 280 = 900\,\mathrm{g}$

Average = $\frac{900}{3} = 300\,\mathrm{g}$

The student is incorrect. The average weight is 300 g.

2.

$$\text{average pages} = \frac{45+60+55+40+50}{5} = \frac{250}{5} = 50$$